Hedging Bitcoin with Its Own Fear Gauge

Evidence from the BVIV and STRC Perpetuals on Volatility Forecasts, Funding Costs and 15-Minute VWAP Timing

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October 2026

Abstract

A perpetual future on the Volmex BVIV index of Bitcoin implied volatility has traded on Bitfinex since April 2024. With 15-minute data from 1 January 2023 to 3 October 2026, including its complete Bitfinex record, we ask how BTC volatility should be forecast when a 30-day implied index is observable and whether a long-BVIV position, possibly timed by a volume-weighted average price (VWAP) signal, hedges BTC. HAR augmented with implied variance is the best one-day forecast, ahead of HAR in every year, significantly over the full window and in 2025 but not since the perpetual's listing; HAR-IV leads at a week and the bias-corrected index at a month, neither significantly. With rules frozen on simulated training data and executed one bar after the signal at the mark, a minimum-variance BVIV hedge reduced the expected shortfall (ES) of a long-BTC book by 17.9% (95% block-bootstrap CI 6.2–32.4%), mostly on a handful of crash days in 2026. Funding settled at its $\pm$0.25% cap in 41% of 8-hour periods and cost a permanently long contract 24 vol points a year; the perpetual traded a median $71 a day, and on its best days the hedge of a single bitcoin exceeded the market's open interest. A VWAP ratchet added 2.37 points of ES reduction (CI [-0.22, +4.89]), but placebos with the same triggers at shifted dates captured most of it: the gain is mainly extra downside exposure, which holding the ratchet's overlay permanently delivers too, with timing worth +0.34 points. Shorting Strategy's STRC preferred stock reduced ES by 16.2% but paid in the issuer's June 2026 funding squeeze, not in bitcoin crashes, at 16.4% of BTC notional a year. A calibrated simulation with an arbitrage-free perpetual reproduces the best forecast at each horizon and the split between exposure and timing. BVIV perpetuals worked as a crash hedge, but funding, execution speed and capacity, not signal design, decide what that hedge is worth.

Keywords: Bitcoin, implied volatility, BVIV, STRC, perpetual futures, hedging, VWAP. **JEL:** C22, C58, G11, G13, G17.

1 Introduction

A holder of bitcoin, in spot or through a perpetual swap, faces two kinds of risk: an ordinary random walk, and sudden crashes in which liquidity disappears and implied volatility spikes. Futures hedges remove the first and, with it, the position itself; crash protection has required options. The Volmex BVIV index measures 30-day BTC implied volatility in a model-free way (Babaoğlu, 2026), and it has traded as a USDT-margined perpetual on Bitfinex since April 2024 (Godbole, 2024), on gTrade since June 2025 (Godbole, 2025a) and on Hyperliquid since September 2026 (Godbole, 2026). Since June 2026 CME has also listed Bitcoin Volatility futures, settled on its own CME CF Bitcoin Volatility Index rather than on BVIV (CME Group, 2026).

Whether long volatility hedges long BTC depends on the spot-vol correlation, which in BTC is weak and unstable. In BTC, positive return shocks raised volatility more than negative ones before the 2013 crash (Bouri et al., 2017) and, with time-varying strength, for some years after (Takaishi, 2021), as they do for most large cryptocurrencies (Baur and Dimpfl, 2018); the asymmetry has since become small (Takaishi, 2021); the BTC–DVOL correlation flipped positive in early 2023 (Godbole, 2023), while BVIV held near 60 in late 2024 and fell to about 40 during the 2025 leg of the rally (Godbole, 2025b). Equity investors learned that long-volatility hedges look good *ex ante* but are eroded by carry and costs (Whaley, 2013; Alexander et al., 2016; Israelov, 2019), except in genuine crises (Szado, 2009). This raises two questions: what does such a hedge cost on a real crypto venue, and can it be *timed* by a signal every intraday trader watches, price relative to its VWAP?

Contributions. (i) We provide, to our knowledge, the first evidence on hedging BTC with a tradable implied-volatility perpetual, including an exact reconstruction of its funding rule, its realised costs and its capacity. (ii) We compare seven forecasts of BTC realised volatility, three of which use the official BVIV index. (iii) We test VWAP timing with rules frozen on simulated data, single-path block-bootstrap inference and a *timing placebo* that splits a rule's edge into exposure and timing. (iv) A calibrated simulation with an arbitrage-free perpetual

serves as a controlled experiment. (v) We test a credit hedge: shorting Strategy Inc.'s STRC preferred stock on Nasdaq and on Hyperliquid. A pipeline regenerates every number from public data, except one dated order-book snapshot and the STRC perpetual's history (0xArchive).

2 The BVIV index and its perpetual

Index. BVIV is a 30-day constant-maturity implied-volatility index that applies the variance-swap replication of Demeterfi et al. (1999) to Deribit and OKX options, smoothed by an EWMA with a half-life of one to two minutes, and is disseminated every second (Babaoğlu, 2026); we use its 15-minute closes. Deribit's DVOL applies a similar calculation to Deribit options alone (Deribit, 2021). Implied variance exceeds subsequent realised variance on average: Almeida et al. (2024) estimate about 0.14 a year in variance units for 2017–22, consistent with Alexander and Imeraj (2021) and the equity evidence of Carr and Wu (2009).

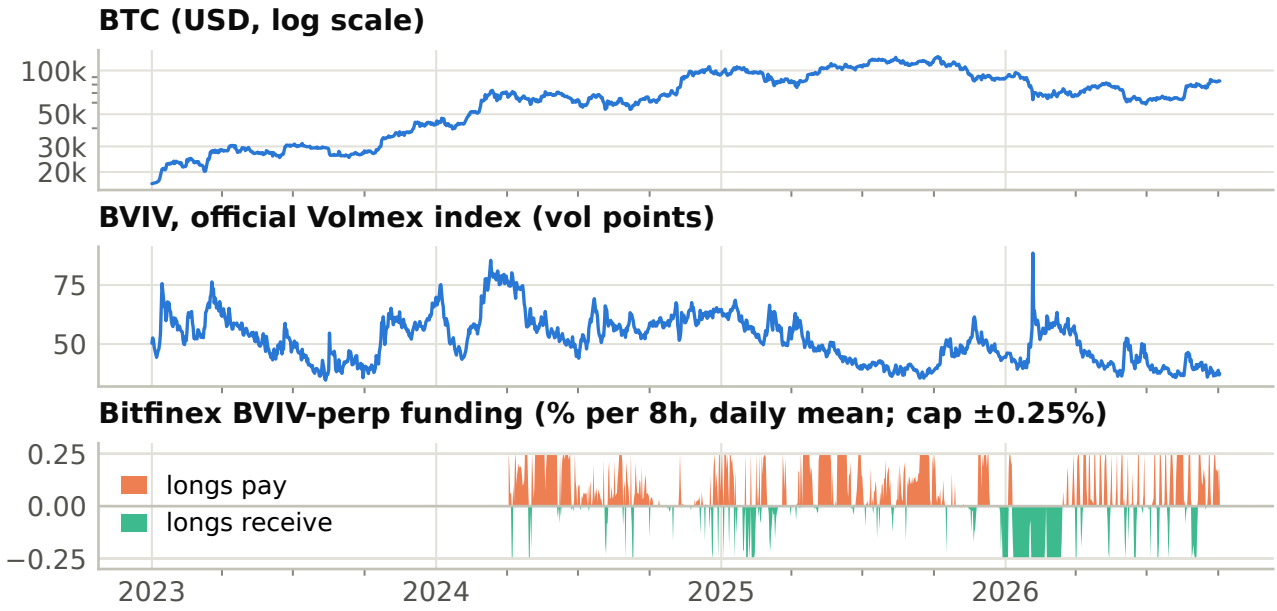


Figure 1: Live data, 1 January 2023 to 3 October 2026. Top: BTC (Binance). Middle: the official Volmex BVIV index. Bottom: Bitfinex BVIV-perpetual funding per 8-hour period (daily mean), paid by longs when positive.

The Bitfinex perpetual. One contract pays one dollar (USDt) per index point; the mark price is the index converted at USD/USDt and deviates from the official Volmex index by at most 0.15% on 95% of 15-minute bars. Funding settles every eight hours. The status feed exposes the accrued premium a of the perpetual over its mark, and the settled rates follow $\text{sign}(a)\min\{\max(|a| - 0.05\%, 0), 0.25\%\}$ exactly (largest deviation 0.05 bp over all 2,740 settlements). The cap, 0.75% of notional a day, binds often: in 41% of settlements from April 2024 to October 2026, and longs paid on 52%. A permanently long contract paid 24 vol points a year on average (51% of notional), but the burden swung from 46 points a year in 2024 (from April) and 34 in 2025 to -12 a year in 2026 (to October): longs *received* 22 points in its first quarter and paid again from April, at 26 points a year (Figure 1). The market is tiny. The perpetual traded on 576 of 914 days, with a median daily volume of $71 (mean $7,300), and only 4.4% of 15-minute bars saw a trade. Its quoted mid sat a median 0.05% above the mark, but dust orders at the top of the book put it more than 5% away on 1.4% of bars (up to +276%), so it is not an executable price.

3 Data and volatility forecasts

Data. BTC is the Binance BTCUSDT spot market on 15-minute bars (1,372 days); quote volume over base volume gives each bar's exact VWAP. BVIV is the official Volmex series from its public 15-minute history. The perpetual's marks, quotes, open interest and accrued funding come from about 1.3 million Bitfinex status snapshots (one a minute), which also yield every funding settlement; trades come from Bitfinex's trade candles. Volatility averaged 46.6% (Table 1) against a mean index of 51.9 (daily closes 34.7–88.5); implied exceeded subsequent realised volatility on 75% of days. The daily spot-vol correlation was -0.10 overall but moved from +0.32 in 2023 through +0.03 and -0.30 to -0.59 in 2026: the market became VIX-like. The index lags BTC: its 15-minute changes are autocorrelated (0.20), and 39% of its response to a 15-minute BTC return arrives in the three following bars, which matters for execution (Section 4).

Moment	Live	2023	2024	2025	2026	Sim.
Realised vol, annualised (%)	46.6	43.2	53.0	44.6	44.3	51.5
Mean BVIV (vol pts)	51.9	52.4	60.2	48.3	45.2	57.1
IV − subsequent 30d RV (vol pts)	6.8					7.6
Share of days IV > RV (%)	75					78
corr(daily return, ΔIV)	−0.10	0.32	0.03	−0.30	−0.59	−0.21
corr on falling 15m bars	−0.24					−0.26
AC(1) of 15m index changes	0.20					–
Worst daily log return (−%)	15.1					15.0

Table 1: Stylised facts: live sample (1 January 2023 to 3 October 2026) and by calendar year, against the median of the simulation's 200 test paths (Section 6).

	$h = 1$ day			$h = 7$ days			$h = 30$ days		
Model	QL ratio	MZ R^2	DM t	QL ratio	MZ R^2	DM t	QL ratio	MZ R^2	DM t
RW	1.71	0.16	10.47	1.33	0.15	2.56	1.42	0.09	1.70
EWMA	1.27	0.09	5.18	1.40	0.10	2.91	1.87	0.09	2.53
GARCH	1.41	0.10	8.96	1.95	0.09	5.29	3.52	0.14	3.21
HAR	1.00	0.21	–	1.00	0.19	–	1.00	0.02	–
IV-raw	1.24	0.15	6.07	1.06	0.22	0.79	0.91	0.26	−0.65
IV	1.17	0.15	2.72	1.02	0.20	0.15	**0.84**	0.18	−1.54
HAR-IV	**0.96**	0.26	−2.25	**0.95**	0.25	−0.74	0.92	0.16	−0.60

Table 2: Live out-of-sample forecast accuracy, 2024 to October 2026. QL ratio: a model's QLIKE over HAR's (best in bold). DM t: Diebold–Mariano statistic against HAR (negative favours the model).

Forecasts. From 15-minute log returns we compute daily realised variance (Andersen et al., 2003); fifteen minutes is coarser than the 5-minute rule of thumb (Liu et al., 2015) but matches the hedge's frequency. Each model forecasts the average daily variance over the next $h \in \{1, 7, 30\}$ days from information at day t's close: a random walk, RiskMetrics EWMA (J.P. Morgan/Reuters, 1996), GARCH(1,1) with Student-t errors (Bollerslev, 1986, 1987) re-estimated monthly, the heterogeneous autoregressive (HAR) model of Corsi (2009) with a 7-day week, the index itself, raw ($(IV_t/100)^2/365$, "IV-raw") and bias-corrected ("IV"), and HAR augmented with implied variance:

$$\ln \overline{RV}_{t+1:t+h} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln \overline{RV}_{t-6:t} + \beta_m \ln \overline{RV}_{t-29:t} + \beta_{iv} \ln\left((IV_t/100)^2/365\right) + \varepsilon_{t+h}. \tag{1}$$

HAR and IV are the special cases without the implied or the realised terms. Regressions run in logs, so forecasts $\exp(\hat{y} + s^2/2)$, with s^2 the residual variance, are positive, and are re-estimated daily on an expanding window of targets already observed. We evaluate the 1,006 one-day forecasts from 2024 on with QLIKE, which is robust to proxy noise (Patton, 2011), the Mincer–Zarnowitz R^2 and Diebold–Mariano (DM) tests against HAR with Newey–West variance (Mincer and Zarnowitz, 1969; Diebold and Mariano, 1995; Newey and West, 1987).

Results. At one day Table 2 agrees with Hoang and Baur (2020) on Deribit data: the index alone loses to HAR and the combination wins. At one day HAR-IV is best (QLIKE ratio 0.96) and ahead of HAR in each year (0.99, 0.93, 0.95), significantly in 2025 (DM −2.66) and over the whole window (DM −2.25) but not from the perpetual's listing on (−1.83), while the index alone is worse (1.17) and GARCH on daily returns lags (1.41), as Bergsli et al. (2022) found for BTC. At longer horizons the leading models (HAR, HAR-IV and the index) are not significantly different. HAR-IV stays ahead at seven days (0.95; DM −0.74), where, unlike in their data, the bias-corrected index is no better than HAR (1.02); at 30 days the bias-corrected index is best (0.84; DM −1.54), followed by raw IV (0.91) and HAR-IV (0.92). The calibrated simulation has the same winner at every horizon (Section 6).

4 Hedging framework

P&L. The book is long q BTC at price S_t and long $H_t \geq 0$ BVIV-perpetual contracts with mark F_t. A position H_t held from the close of 15-minute bar t earns over bar $t+1$

$$\begin{aligned}\Pi_{t+1} = {} & q\,(S_{t+1} - S_t) - \mathbb{1}_{\text{perp}}\, q\, S_t f_t\, \Delta t + H_t\,(F_{t+1} - F_t) - H_t\,\psi_t \\ & - (H_{t+1} - H_t)(P_{t+1} - F_{t+1}) - c\,|H_{t+1} - H_t|\,F_{t+1},\end{aligned} \tag{2}$$

where $\mathbb{1}_{\text{perp}} = 1$ for a perpetual BTC book, Δt is the bar length in years, f_t the BTC perpetual's annualised funding rate, ψ_t the BVIV funding per contract, booked on the position held at each settlement, P_t the fill price,

and c a proportional cost of 6 bp fee plus 10 bp slippage per side. On live data ψ_t is the settled funding. A perpetual on an index I_t is priced at the expected index sampled at a random time (He et al., 2022; Ackerer et al., 2026); in the simulation we split its funding into an arbitrage-free drift transfer $\mathbb{E}_t[I_{t+1}] - I_t$ and a carry premium.

Execution. On live data a decision at the close of bar t is executed at the close of bar $t+1$ at the mark ($P = F$), plus costs. Same-bar execution would collect the index's predictable catch-up after a BTC move (Section 3), and the quoted mid is not executable (Section 2); both are sensitivities (quotes capped at ±5% of the mark), as are costs of 50 bp per side on top of the fee, the price impact of one bitcoin's hedge in the order book a day after the sample. No position is held before the listing at 09:00 UTC on 3 April 2024.

Hedge ratios and signal. The minimum-variance hedge is $H_t^* = -qS_t\beta_t$, with $\beta_t = \mathrm{Cov}_t(r, \Delta F)/\mathrm{Var}_t(\Delta F)$ estimated on 15-minute BTC log returns r and mark changes ΔF by an EWMA (half-life 14 days) (Johnson, 1960; Ederington, 1979); we impose $H \geq 0$. A downside ratio $\beta_t^- = \mathbb{E}_t[r\,\Delta F \mid r < 0]/\mathbb{E}_t[\Delta F^2 \mid r < 0]$ on falling bars, shrunk towards β_t with a two-day prior, gives the crash hedge H_t^-. The breakdown statistic compares the price with its rolling 24-hour VWAP,

$$z_t = \ln\big(S_t/\mathrm{VWAP}_t^{24h}\big)\big/\big(\hat{\sigma}_t\sqrt{1/3}\big), \tag{3}$$

because for Brownian prices the gap between the last price and its average over a window w has standard deviation $\sigma\sqrt{w/3}$. Here $\hat{\sigma}_t$ is the daily volatility implied by a causal EWMA (half-life one day) of 15-minute squared returns, de-seasonalised by a trailing hour-of-week profile (Andersen and Bollerslev, 1997; Hansen et al., 2024).

Rules. All rules are long-only, their targets are capped at 50% of BTC notional, and they trade only when the gap between target and position exceeds 25% of the larger of the two, or when a switch flips or the target drops to zero (Whalley and Wilmott, 1997), so held positions can drift above the cap (on live data at most 64% for the minimum-variance hedge and 72% for 1.5×). *Always-on* holds λH_t^* with $\lambda \in \{1, 1.5\}$. The *VWAP-switch* holds H_t^- while a gate is on, switching on at $z_t < z_{in}$ and off at $z_t > z_{out}$, for at least four hours. The *VWAP-ratchet* holds a core φH_t^* plus an overlay o_t that jumps to 1 on a breakdown and decays with half-life τ:

$$o_t = \begin{cases} 1 & z_t < z_{in} \\ 2^{-1/(96\tau)}\, o_{t-1} & \text{otherwise,} \end{cases} \qquad H_t = \varphi H_t^* + o_t \max\big(H_t^-\, m_t - \varphi H_t^*,\, 0\big), \tag{4}$$

where $m_t = \min\{\max(e^{\kappa_t}, 0.5), 1.5\}$ optionally scales the overlay by forecast "cheapness" κ_t: the log ratio of the 30-day HAR-IV forecast to implied variance, less its trailing 180-day median, lagged one day.

Protocol. Rule parameters were selected on 48 simulated training paths (Section 6), out of 72 ratchets and six switches, by median ES reduction net of costs, before the live study was run. The selected ratchet keeps the full core ($\varphi = 1$), triggers at $z < -1.5$, decays with a half-life of 4 days and uses forecast sizing; the selected switch enters at $z < -1$ and exits at $z > 0.5$. The live sample is therefore out of sample for the rules, which avoids data snooping (White, 2000; Hansen, 2005; Bailey et al., 2014), although the simulator's calibration targets include coarse features of the 2023–26 market. The simulation, including the selection, fills on the signal bar, as its index has no lag; there the selected ratchet led the next by only 0.02 points, and one-bar fills would have picked $z_{in} = -1$ (live timing value +0.11). Signals warm up on 2023 data, and hedging is evaluated over the 914 days since the listing, 3 April 2024 to 3 October 2026. On a single path we report 95% intervals from a paired stationary block bootstrap (Politis and Romano, 1994) over days (mean block ten days, 2,000 draws); for seven mean block lengths from 1 to 40 days the lower bound of the always-on ES reduction lies between 4.4% and 7.3%. The timing placebo shifts a ratchet's trigger series circularly within the whole weeks of the evaluation window, by every multiple of a week for the selected rule (129 shifts) and of two weeks for each rule of the grid (64 shifts). It keeps the triggers' number, clustering and hour-of-week profile but breaks their link to the market. A ratchet's edge over the always-on hedge then splits into *exposure*, the placebo mean minus the always-on result, and *timing*, the ratchet minus its placebo mean.

5 Results on live data

The hedge works, in crashes. The minimum-variance hedge reduced ES by 17.9% (block-bootstrap CI 6.2–32.4%) and variance by 14.9% (3.9–28.0%), and cut the maximum drawdown from 53.0% to 41.7% with an average notional of 11.1% of the BTC position (Table 3). It was held on 67% of bars and off whenever the estimated spot-vol covariance was positive, including on 8 of the 23 days in the unhedged ES tail. Net of funding (Figure 2), its P&L summed to +5.3% of BTC notional: the ten best days contributed +51% of notional

Rule	ES red.	Var red.	Max DD	Tail offset	Hedge P&L	Funding	Trading	Turnover	Notional
Unhedged	0.0	0.0	53.0	0.0	0.00	0.00	0.00	0.0	0.0
Always-on (MV)	17.9	14.9	41.7	26.9	2.13	1.41	0.20	1.3	11.1
Always-on, 1.5×	20.1	13.2	41.9	38.7	1.07	2.21	0.29	1.8	16.7
VWAP-switch	18.8	12.6	42.5	25.6	6.71	−0.26	5.17	32.3	4.6
VWAP-ratchet	20.3	15.0	42.6	30.6	1.35	2.67	0.37	2.3	12.5

Table 3: Hedging a 1-BTC spot book with the Bitfinex BVIV perpetual, 3 April 2024 to 3 October 2026, executed one bar after the signal at the mark (%, except turnover). ES/Var red.: reduction in daily $ES_{97.5}$ and variance against unhedged. Max DD: maximum drawdown of compounded daily returns. Tail offset: hedge P&L net of costs as a share of BTC losses on the worst 5% of days. Hedge P&L (index P&L minus funding), Funding and Trading (fees and slippage): % of BTC notional a year. Turnover: perpetual notional traded per year over BTC notional. Notional: average hedge notional in % of BTC.

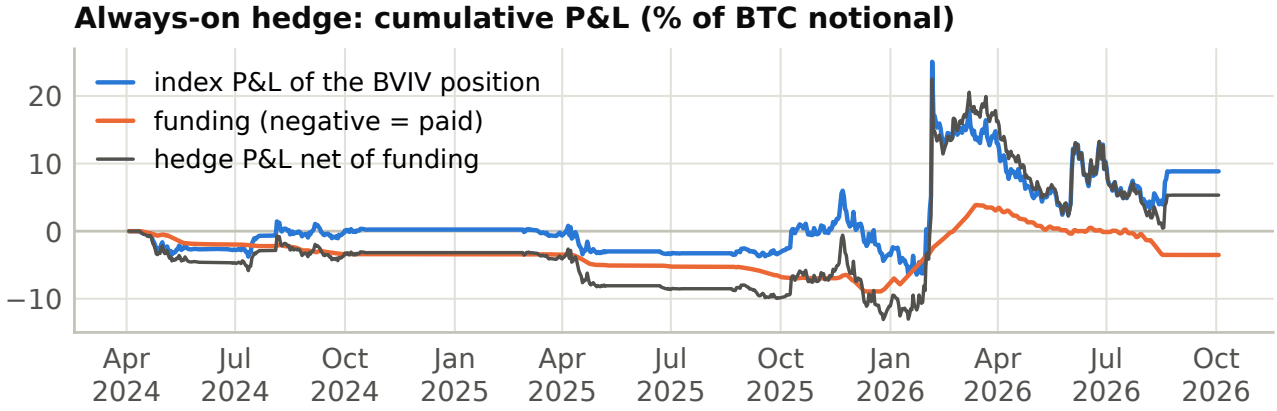


Figure 2: Always-on minimum-variance hedge on live data: cumulative index P&L of the BVIV position, cumulative funding (negative = paid), and hedge P&L net of funding, in % of BTC notional.

Variant	Always-on	1.5×	Switch	Ratchet
Same-bar execution	19.1 (+2.5)	20.5 (+1.5)	20.1 (+7.6)	21.1 (+2.7)
Execution 1 hour after the signal	16.0 (+0.4)	17.3 (−1.9)	14.1 (−3.0)	18.3 (−2.0)
Execution 1 day after the signal	12.6 (−1.4)	11.2 (−5.3)	10.3 (−4.3)	13.2 (−3.0)
Fills at the quoted mid	17.7 (+1.6)	19.8 (+0.4)	18.4 (+2.8)	20.2 (+0.4)
Book costs (50 bp + fee)	17.8 (+1.4)	20.0 (+0.1)	18.2 (−11.4)	20.1 (+0.1)
No trading costs	18.0 (+2.1)	20.1 (+1.1)	19.0 (+6.7)	20.3 (+1.4)
Funding ignored	17.9 (+3.3)	20.0 (+3.0)	18.7 (+1.3)	20.3 (+3.6)
Perpetual BTC book	18.0 (+1.9)	20.1 (+0.8)	18.9 (+1.5)	20.3 (+1.0)

Table 4: Live-data sensitivities, frozen rules, one change at a time to the baseline (execution one bar after the signal at the mark, 6+10 bp per side): ES reduction in % (hedge P&L net of funding and trading costs, % of BTC notional a year).

and all other days −46%. The ten worst BTC days cost the unhedged book 74% of notional, and the hedge earned back 36%, 48% of that loss. The ES reduction was 6.3% in 2024, 10.0% in 2025 and 52.3% in 2026; without the best day it is 11.9%, without the five best 9.0%. Over 7- and 14-day horizons it is 22.2% and 32.4%, so the daily figure is not an artefact of the index lagging across day boundaries. Funding cost the hedge 1.4% of BTC notional a year: it paid 4.6% a year in 2024 and 4.3% in 2025 and received 5.6% in 2026.

The best day. On 5 February 2026 the hedge returned 20% of BTC notional while BTC fell by 14% (Figure 3). BVIV stood at 57 at midnight and 61 an hour before BTC broke below the ratchet's VWAP band on the bar ending 15:15 UTC; it was 67 on that bar, 77 by 15:45 and peaked at 96 at 22:00. All rules entered the day with 558–609 contracts. As the index's 15-minute moves exploded, the EWMA hedge ratio fell and the rules sold into the spike, the always-on hedge to 205 contracts by the close; the ratchet's overlay kept it up to 20% larger.

Timing adds little beyond exposure. The switch traded 32.3 times the BTC notional a year; its ES reduction differed from the always-on hedge's by +0.92 points, not significantly (CI [−2.71, +3.54]), and at the book's prices it would have paid 18.1% of notional a year in trading costs. The ratchet beat the always-on hedge by +2.37 points (CI [−0.22, +4.89]) and the 1.5× hedge by +0.17 (CI [−1.74, +2.35]). Its 129 placebos reduced ES by 19.9% on average, 90% of them between 19.1% and 20.4% (Figure 4, left), so +2.03 points of the edge are exposure and +0.34 timing; the ratchet beats 87% of its placebos (98% with same-bar fills, 71% at quoted

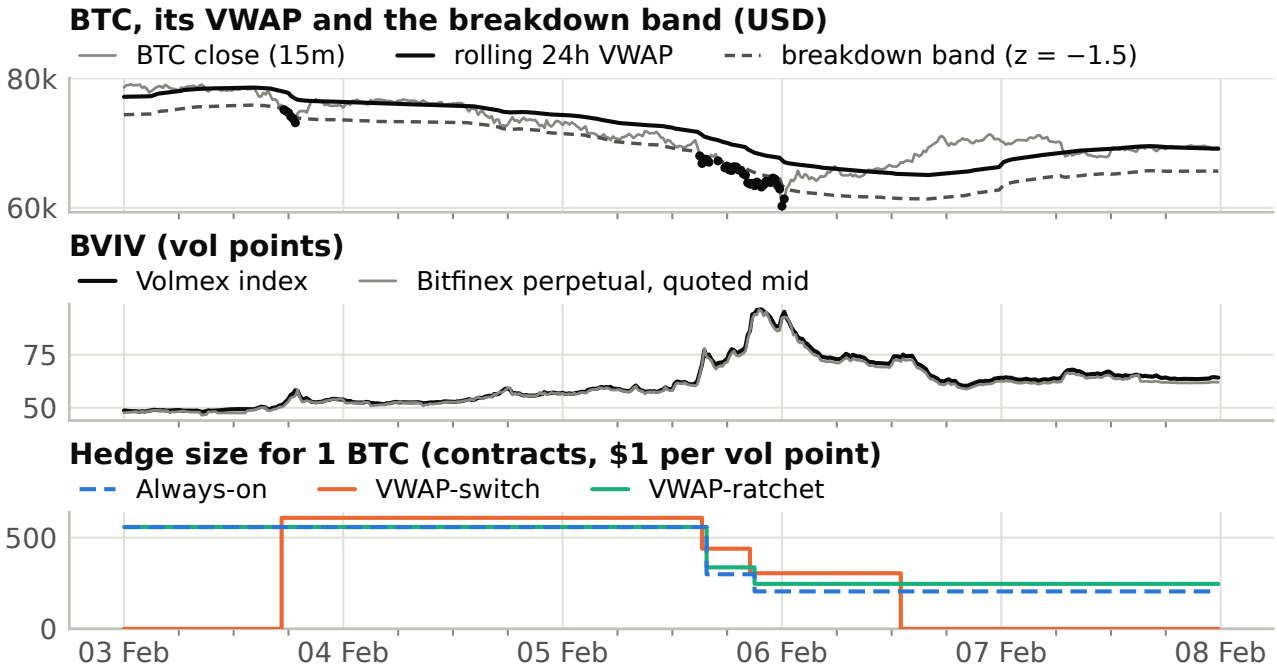


Figure 3: The best hedge day, 5 February 2026, and the days around it at 15-minute resolution (UTC, bar open times). Top: BTC, its rolling 24-hour VWAP and the ratchet's breakdown band (dots: $z_t < z_{in}$). Middle: the Volmex index and the Bitfinex perpetual's quoted mid. Bottom: hedge sizes of the three rules for a 1-BTC book.

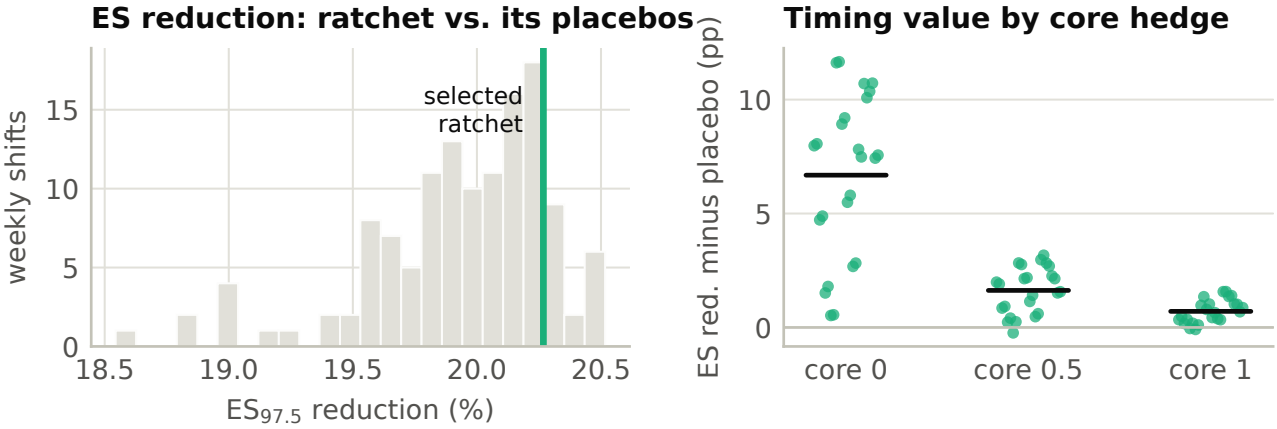


Figure 4: Timing placebos on live data. Left: ES reduction of the selected ratchet (line) against its 129 weekly circular shifts of its triggers. Right: timing value of each of the 72 ratchet configurations (ES reduction minus the mean over 64 two-weekly shifts) by core size; bars mark the means.

mids). Notional explains only part of the exposure term on point estimates: scaled to the placebos' notional (1.13×), the minimum-variance hedge reduces ES by 18.6%, 1.3 points less than the placebos, not significantly (CI [−0.73, +3.01]), whereas the ratchet's overlay held permanently reduces it by 20.5%, as much as the best timed rule (20.4%; +0.26 points against the selected ratchet, CI [−0.94, +1.81]), for 3.65% of notional a year against the ratchet's 3.03%. Across the 72-rule grid, timing adds +6.68 points without a core hedge (rules beat a mean 97% of their placebos), +1.63 with half a core and +0.71 with the full core (83%; 10 of 24 rules beat at least 95%; right panel); for the strictest full-core rules ($z_{in} = -2$, half-lives up to two days) timing is 51–84% of the edge. VWAP breakdowns time protection far better than chance when nothing else is held, but they flag a sell-off already under way, and a core hedge captures most of their value.

Execution and capacity. Costs, funding and fills at quoted mids barely move the ES reductions, which are dominated by crash days (Table 4). Speed matters most: same-bar execution would raise the always-on ES reduction to 19.1%, while executing one hour or one day after the signal lowers it to 16.0% and 12.6%. Capacity binds. When held, the hedge of one bitcoin was a median 248 contracts, against a median open interest of 103 on those bars, and exceeded the whole open interest on 61% of those bars (41% of all bars). On its five best days its peak was 1.7 to 23 times the day's median open interest; on 5 February 2026 it traded 353 contracts while the market traded 18, and 30% of its trades fell on days without a single trade in the market.

Rule	ES red.	Var red.	Max DD	Tail offset	Carry	Trading	Turnover	Time on
Always-on (MV)	4.2	4.1	48.7	5.9	0.74	0.14	0.9	93
Always-on, 1.5×	4.9	3.1	49.5	8.8	1.11	0.20	1.3	93
VWAP-switch	2.8	1.9	50.9	4.9	0.38	3.94	24.6	35
VWAP-ratchet	4.9	3.9	48.9	7.8	0.96	0.23	1.4	98
Ratchet, placebo timing	4.9	3.8	48.7	7.7	0.96	0.24	1.5	98

Table 5: Simulation: hedging a long-BTC book on 200 test paths (medians; % unless stated). Columns as in Table 3, but with fills on the signal bar; Carry: funding carry paid (% of BTC notional a year); Time on: share of bars hedged. Placebo timing: the selected ratchet with circularly shifted triggers (mean of 5 shifts).

A book snapshot on 4 October 2026 shows market makers quoting size: one bitcoin's hedge would have cost about 0.50% per side, half the 1.00% spread between the first levels of at least one contract (the top of the book held dust), and depth within 2.5% of the mid would have held the hedge of about 4 bitcoins.

6 The controlled experiment

One path cannot separate rules reliably, so we repeat the study in a simulated market calibrated to the stylised facts (Table 1, last column). Log-variance is the sum of a slow and a fast Ornstein–Uhlenbeck factor; the fast factor tracks a three-state hidden regime (calm, stress, euphoria) and receives EGARCH-type shocks (Nelson, 1991) whose signed loading on the return shock is $-0.30/-0.60/+0.25$ by regime; jumps cluster in stress, and volatility and volume are seasonal within the day and week. The index is the square root of the model's closed-form expectation of 30-day average variance (a Feynman–Kac product over regime paths, lognormal in the factor shocks), scaled by a variance premium and perturbed by persistent sentiment. The perpetual's funding is the one-bar drift transfer (exact over regimes and jumps, second order in the diffusive shocks) plus a carry of 6 vol points a year, so it offers no free lunch from mean reversion; unit tests check both expectations against brute-force simulation. The simulated index has neither the live index's lag nor a funding cap, so the simulation does not speak to execution delay or the funding regime. Rules are selected on 48 training paths of 900 days and evaluated on 200 test paths.

Findings. The best forecasts are the same as on live data: HAR-IV is best at one day (median QLIKE ratio 0.95) and seven days (0.89), the bias-corrected index at 30 days (0.85). The hedging gains are smaller (Table 5): the simulation has neither the 2026 crash cluster nor its strong spot-vol correlation (-0.59 against -0.21). The always-on hedge reduces ES by a median 4.2%. Averaged over paths, the paired difference to it is -1.67 points for the switch (CI $[-2.00, -1.36]$), unlike on live data, where the two cannot be told apart, and +0.53 for the ratchet (CI [+0.33, +0.72]), whose edge over its placebo is only +0.05, and a 1.5× hedge reaches the same ES reduction (4.9%): as on live data, the edge is mostly exposure. Across the grid (64 paths, 3 shifts per rule), timing adds +0.78 points without a core and +0.02 with the full core. In 96 further paths with common random numbers, where its edge is +0.66 points (CI [+0.39, +0.94]), it survives doubled or halved trading costs and a carry of 0 to 24 vol points a year (+0.63 to +0.67 points). It shrinks to +0.18 points under inverse leverage and to +0.34 when crashes cluster in stress, where the CIs include zero.

7 A credit hedge: short STRC

The instrument. STRC ("Stretch") is the variable-rate perpetual preferred stock of Strategy Inc., whose assets are mostly bitcoin (Strategy Inc., 2025). Sold at $90 in July 2025, it has traded on Nasdaq since 30 July 2025. Its $100 stated amount earns a cumulative cash dividend whose rate Strategy adjusts to keep the price near $100, and Strategy may redeem it at $101. The rate rose from 9% to 11.5% by March 2026 and to 12% from July 2026; since 30 June 2026 the dividend has been paid twice a month (Strategy Inc., 2026). STRC ranks behind Strategy's debt and its STRF preferred but ahead of its other preferreds and its common stock. A senior claim on a bitcoin balance sheet is riskless debt minus a put on the issuer's assets (Merton, 1974), so a short position in STRC is long that put: it should pay when bitcoin falls far enough to threaten the dividend, and its carry, the dividend, is the premium for selling such catastrophe insurance (Coval et al., 2009). Since 22 June 2026 a perpetual on one STRC share, xyz:STRC, has traded on Hyperliquid, deployed by trade[XYZ] under HIP-3 (trade[XYZ], 2026). Its oracle is the external STRC price whenever a US session, including an overnight ATS session, is open, and an internal order-book average at weekends, inside ±10% discovery bounds. It is not adjusted for dividends: a short collects the ex-dividend drop and pays the perpetual's hourly funding instead of the dividend.

Design. The frozen rules of Section 4 are applied unchanged, but the hedge is short: the EWMA slope of BTC returns on STRC price changes sets the size whenever it is positive. Nasdaq prices exist at hourly closes

Hedge	ES red.	Var red.	Max DD	Hedge P&L	Carry	Trading	Notional
Short STRC on Nasdaq, frozen rules							
Always-on (MV)	16.2	22.4	57.3	−13.8	15.9	0.49	127
Always-on, 1.5×	10.6	20.0	59.9	−19.2	22.0	0.49	178
VWAP-switch	13.8	16.5	57.7	−4.6	3.7	5.88	48
VWAP-ratchet	15.5	22.2	57.4	−13.5	16.8	0.57	129
Always-on hedges, same trading days							
Long BVIV perpetual	47.7	33.0	42.2	12.3	−1.6	0.25	21
Half of each	38.6	35.3	48.0	−0.7	7.1	0.37	74
From the perpetual's listing (71 trading days), same hedge ratio							
Short STRC on Nasdaq	2.1	10.3	9.1	−22.7	9.7	0.18	70
Short xyz:STRC perpetual	11.4	14.8	8.8	−20.7	6.3	0.52	70

Table 6: Hedging a 1-BTC spot book by shorting STRC, 29 August 2025 to 2 October 2026 (275 US trading days, % except Notional). Columns as in Table 3; Carry: dividends and borrow fee (Nasdaq) or funding (perpetual), % of BTC notional a year. Unhedged maximum drawdown: 53.1%. The middle panel holds the BVIV hedge of Section 5 on the same trading days, alone and at half size next to half the STRC hedge; the bottom panel runs the same hedge ratio on Nasdaq and on the perpetual from the latter's listing.

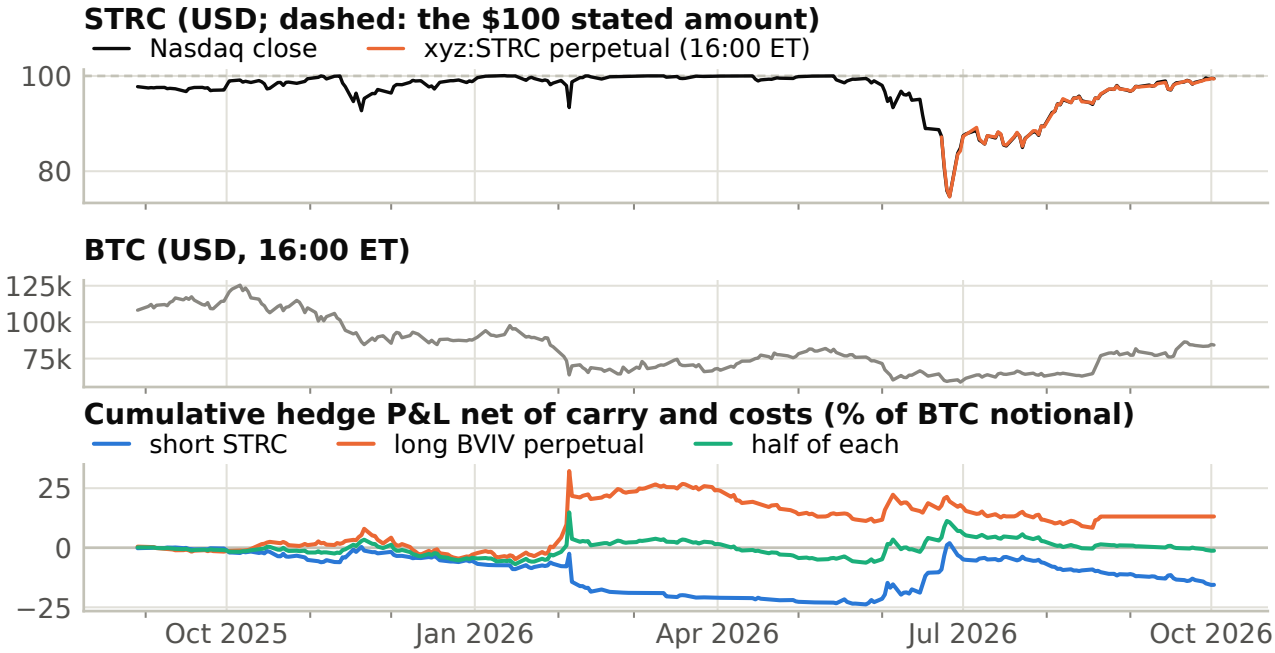


Figure 5: STRC on Nasdaq and on the Hyperliquid perpetual, BTC, and the cumulative P&L, net of carry and trading costs, of the always-on short STRC hedge, the always-on long BVIV hedge and half of each, over US trading days.

in regular hours, so the slope uses only synchronous moves, BTC returns accumulated over the same intervals, and a decision is filled at the next hourly close; on the perpetual, at the next 15-minute mark. A Nasdaq short pays every dividend and a borrow fee of 1.00% a year, and 1+2 bp per side; a perpetual short pays funding and the 9 bp taker fee. The notional cap carries the BVIV study's risk budget over: half the BTC notional, scaled by how much more a dollar of the BVIV perpetual moved than a dollar of STRC over STRC's first 30 days (5.3×), gives 2.65 times the BTC notional. Hedging runs from 29 August 2025 to 2 October 2026, 275 US trading days, and books are compared over trading days, so that a weekend BTC move and STRC's Monday gap fall on the same day. The bootstrap and the weekly placebos are those of Section 4.

A hedge against the issuer, not the crash. Shorting STRC reduced ES by 16.2% (CI [+1.4, +25.8]) and variance by 22.4% (CI [+10.8, +34.0]), but it deepened the maximum drawdown to 57.3% and cost 16.4% of BTC notional a year, almost all of it dividends, at an average notional of 127% of the BTC position (Table 6). STRC is a weak BTC hedge: its daily correlation with BTC was 0.40 and its beta 0.19; on BTC's worst 5% of days BTC fell 5.9% and STRC only 1.3% on average. The issuer's rate policy pulled it back to par. In the crash of 5 February 2026 STRC fell 4.7% and regained 5.8% the next day, after the jump in the EWMA slope had enlarged the hedge, so over that week the hedge lost 7.6% of BTC notional while BTC fell 15.5%. It paid instead in June 2026, when a squeeze on Strategy's funding drove STRC to 74.68 on 26 June 2026 (Van Straten, 2026): a cash reserve

depleted by a convertible buyback, first sales of bitcoin to pay preferred dividends, de-pegging stablecoins backed by STRC and a market value below the bitcoin held, the channel from a bitcoin drawdown through preferred funding to forced sales (Wen, 2026). Over the 9 sessions from 15 June the hedge earned 19.4% of BTC notional while BTC lost 5.9%, and it gave back much of that when a new capital framework and a 12% rate lifted STRC by 12% on 29 June. Four of the five best hedge days fell in June 2026; without the five, the ES reduction is 8.3%. Timing did not help: the ratchet differed from the always-on hedge by -0.7 points (CI $[-3.3, +1.2]$) and beat 16% of its 56 placebos, the switch by -2.5 (CI $[-5.8, +0.6]$). Measured over UTC days, which split weekend BTC moves from the Monday gap, the ES reduction is 14.1%; with a cap of one BTC notional, 12.4%.

The perpetual and the comparison with BVIV. A perpetual short paid 5.3% of notional a year in funding on average (longs paid in the late-June crash, at 21% annualised; shorts paid 18% in September), against 12% in dividends for a Nasdaq short. From each Nasdaq close to the next open, the perpetual's move tracked the opening gap with correlation 0.92 (slope 0.96, 71 nights; 0.98 over 14 weekends), so a weekend move can be hedged when it happens; off-hours prices carry information even when trading is thin (Barclay and Hendershott, 2003). Over the 71 sessions from 24 June 2026, the same hedge reduced variance by 14.8% on the perpetual against 10.3% on Nasdaq (difference CI [+0.6, +9.8]), at 6.8% against 9.8% of BTC notional a year. Capacity is ample: STRC traded a median \$129m a day on Nasdaq, and on the perpetual one bitcoin's hedge, a median \$32k, was 0.34% of open interest (median \$10m), and selling it at the US close cost a median 13 bp against the mid (90th percentile 41 bp). The risk lies in the venue: STRC is margined in isolation without a backstop liquidator, so a profitable short is first in line for auto-deleveraging exactly when the hedge pays (trade[XYZ], 2026; Chitra, 2025). On the same trading days the BVIV hedge reduced ES by 47.7% (STRC minus BVIV: CI $[-42.0, -18.0]$) and variance by 33.0% at a carry of -1.4% a year (Figure 5). The two hedges' daily P&L correlated at 0.56. Half of each reduced ES by 38.6%, less than BVIV alone (CI $[-13.7, -3.2]$), but variance by 35.3%, the most of any hedge. Volatility and credit protect against different events: BVIV against the crash, STRC against distress at the issuer.

8 Limitations and conclusion

The live sample is one path of 914 days (275 trading days with a single credit episode for STRC), and the hedge's gains rest on a few crashes in 2026 (without the five best hedge days the ES reduction falls from 17.9% to 9.0%), which the bootstrap quantifies but cannot remove. Fills at the mark plus costs are a benchmark for small books: on its best days the market could not have absorbed even a 1-BTC hedge, and we observe the order book only once, after the sample. The gains also assume solvent shorts through a 70% intraday rise of the index and a stable USDt; we model neither margin, liquidation and auto-deleveraging nor venue failure. Oracle risk (Hyperliquid, 2026) and algorithmic flow at the 15-minute boundary (Kim and Hansen, 2026) argue for working orders inside the bar by VWAP slicing (Berkowitz et al., 1988; Konishi, 2002; Białkowski et al., 2008).

Five lessons follow. (1) HAR augmented with implied variance is the best one-day BTC volatility forecast, significantly so over the full window and in 2025 but not since the perpetual's listing; at a week HAR-IV and at a month the bias-corrected index lead HAR, not significantly. (2) A minimum-variance BVIV hedge cut tail risk by 17.9% over 2024–26, mostly through a few crashes with spiking implied volatility in 2026. Holding beats switching on costs and in the simulation; on live data the two cannot be told apart. (3) Its price is funding, which on Bitfinex is capped, volatile and on average paid by longs (24 vol points a year), although they received it in early 2026; its limits are execution speed and capacity. (4) A 15-minute VWAP breakdown is real but secondary information: without a core hedge it beats random timing, but with one, on average 66% of a ratchet's edge is extra exposure, which holding its overlay permanently delivers too; timing adds +0.34 points for the frozen rule and +0.71 across the full-core grid; intraday timing rules should face such placebos before they earn their complexity. (5) Shorting STRC hedges the issuer rather than the crash: it paid in Strategy's funding squeeze of June 2026, not when bitcoin crashed in February; its dividend makes it costly on Nasdaq, and its perpetual is cheaper, trades around the clock and is deep but exposes a profitable short to auto-deleveraging.

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